

DAVID CONNER

Software engineer with experience in networking, databases, containers, and web applications. Seeking role that leverages software to solve domain-specific problems, whether as an application programmer or devops. Experience integrating disparate systems with minimal service interruptions and no loss of data.

Fast learner with a thirst for staying ahead in technology world. Prefers the source code as the canonical documentation for most tools, but also reads the docs.

Cisco: **CCNA** (2007, exp)

Microsoft: **MCITP: DBA** (2008, exp); **MCITP: DBD** (2008, exp)

Skills & Tools

Linux	Docker, Podman, Systemd, RPM, Guix, NixOS, LVM, Qemu, KVM
Security	Firewalls, GPG, PIV, Yubikey, X.509, SOPS, AGE, Crypto
Homelab	Cisco, VyOS, VLANs, NAT, Bare Metal, Proxmox
Cloud	Terraform, k8s, GCP, DNS
Lang	Ruby, Python, Java, JS, TS, Clojure, Scheme, Julia, Swift, Lua
Tools	Bash, Ansible, Emacs, SSH & Tunnels, Tmux, Make, Git & Worktrees, Jupyter, AMD ROCm
HW	Robotics, VTY Console, GNU Screen
Data	Reporting, ETL, jq, JSON & Schemas, Postgres, MSSQL, SQLite3, Parquet, graphviz

Formal Education

Experience

Virginia Western *Eng. Student Aide* Roanoke, VA **2022-2023**

- › Maintained Ender-3 Pro and Raise3D printers. Synced Ender-3 configurations for PLA plastics
- › Created an Autodesk Fusion CNC config for a Velocity CNC
- › Collected notes on almost all equipment including support links and digital copies of manuals

RAKE Digital *Cloud Engineer* Salem, VA **2018**

- › Used MS SQL table metadata to quickly learn the accounting database schema for Millennium and ReadyPay
- › Designed an application stack with LoopbackJS and Angular 6 to automate payroll tasks in Azure

Voxxel *Founder* Roanoke, VA **2015-2017**

- › Voxxel enabled fans to score their impersonations of movie quotes and accents
- › Built a Rails API to back prototypes in iOS, Android and AngularJS. Each client processed and visualized the FFT

Oscil8 *Founder* Roanoke, VA **2011-2015**

- › Oscil8 was designed to be the "Github for Music Producers"
- › Developed a business model and strategic vision

Left+Right *Software Developer* Contract, Remote **2014**

- › Developed a web service to extend a Ruby on Rails application with reporting on SQL Views
- › Cached report results in MongoDB to enable a dashboard

Jumpcloud *Software Engineer* Boulder, CO **2013**

- › Full stack development using a NodeJS API and MongoDB
- › Integration tests using Mocha, Selenium and Soda

Weedmaps *Software Engineer* Denver, CO **2012-2013**

Sic Semper LLC *Owner, Developer* Contract, Remote **2011-2012**

Internet Marketing Ninjas *Ruby Developer* Albany, NY **2012**

ABS *Database Developer* Virginia Beach, VA **2011**

CCS *C# Developer* Christiansburg, VA **2010-2011**

ABS *Network Associate* Roanoke, VA **2007-2010**

Virginia Western *Mechatronics S.E.T.* **2006-2008, 2021-2023**

- › Obtained a Cisco CCNA certification
- › Studied design, manufacture, electronics, CNC and safety
- › Planning one course per semester to utilize the Fab Lab

Virginia Tech *Computer Science* **2004-2006; 2008**

- › Studied Computer Science. Dropped out to compete in jam skating at a national level

Continuing Education

Coursera *Perennial Education* **2012-2016**

- › Certificates: Epigenetics, 2014; Bioinformatics I/II, 2015
- › Other: Machine Learning, 2012; Drugs in the Brain, 2014

Self-Directed Study *Perennial Education* **2015 -**

- › Watched at least 1,000 hours of YouTube lectures on mathematics, engineering and emerging fields.
- › Used the zettelkasten method to synthesize insights from dozens of fields. Wrote essays combining cybernetics, semiotics, artificial intelligence, agency and sociology
- › Designed a graphics library for Swift to leverage functional composition for dynamic rendering pipelines using features unique to Metal

Homelab *Automation* **2021-**

- › Developed ansible playbooks SDN for VLANs, Firewalls and IP Migration
- › Created a Guix System image for GPG and Smallstep CA

Workshop *Craftsmanship* **2021-**

- › Modified online designs to build a workbench and shelving
- › Organized a workshop for woodworking, electronics, and making art supplies

Interests Math, 3D Graphics, 3D Design, Philosophy, Futurism, Writing, Linguistics, Semiotics, Bioinformatics, Epigenetics, Colorimetry, Logistics, Materials

Hobbies FOSS, Learning, Painting, Drawing, Kaggle, Electronics, Dance, Board Games, Writing

Apps Blender, Krita, FreeCAD, OpenCascade, Inventor, Matlab, IRC

Volunteer Experience

FRC Team #10004 *Programming Mentor* Roanoke, VA **2024-2026**

- › Mentored talented high-schoolers to collaborate with Java & Git
- › Converted CAD assets to STEP files and helped to program a digital twin with simulated control systems

Make Roanoke *Volunteer* **2023-2026**

- › Worked on an Astro.js static website & built an Astro-native component design system (2024). A simpler framework was selected.